

$$\begin{array}{ccc}
 U' & \xrightarrow{f} & f(U') \\
 \downarrow & & \downarrow \\
 A_r & \xrightarrow{\quad} & A_R \\
 \downarrow h & & \downarrow \widetilde{h} \\
 A_r & \xrightarrow{z^d} & A_R
 \end{array}$$